

Kestrel Bank

AML Detection

Find 4.4x More Suspicious Cases Without Hiring More Investigators

PRESENTED BY

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Adaptive Alert Ranking

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WHAT WE SHOULD DO

Replace the outdated rules based alert ranking with an AI powered Machine learning model that uses account behavior, network relationships, and monthly retraining to prioritize suspicious activity.

ADAPTIVE ALERT MODEL ENHANCES DETECTION

716 CASES IDENTIFIED

Identified 10.44% of laundering using 50 reviews per day

TRADITIONAL RULES BASED MODEL DETECTS SIGNIFICANTLY LESS

163 CASES IDENTIFIED

Identified 2.38% of laundering using 50 reviews per day

Why this matters

FINANCIAL PENALTIES AND COMPLIANCE COSTS

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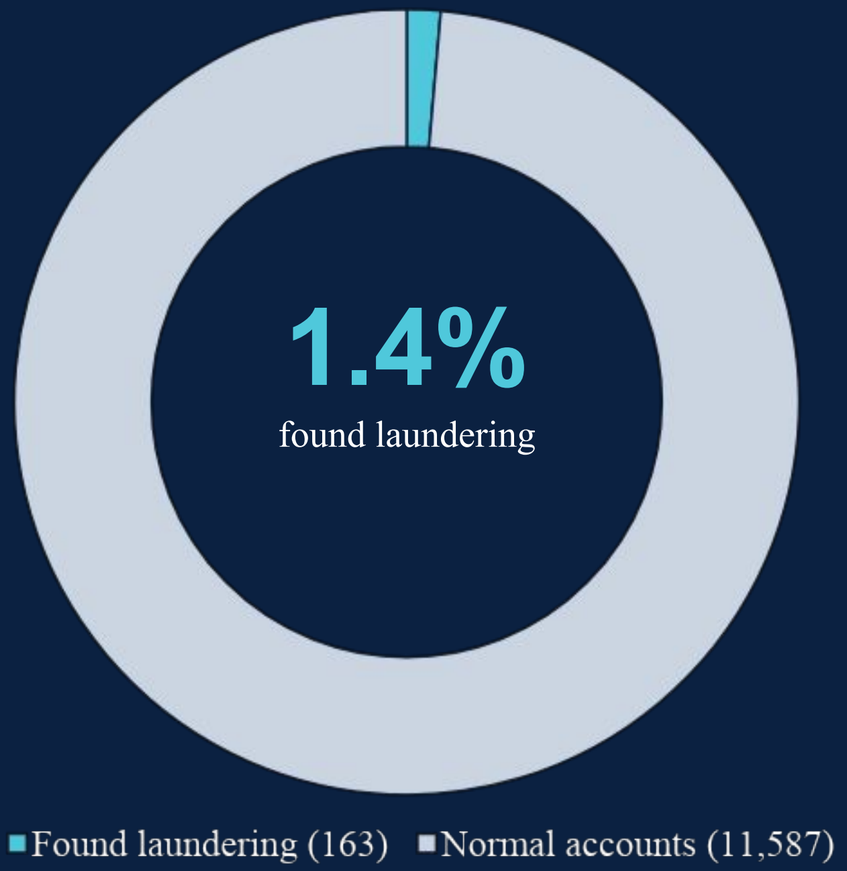


- Analyst labor is the largest cost in financial crime compliance, about $61B a year across U.S. and Canadian banks
- Weak monitoring is a regulatory exposure: TD Bank paid about $3B in October 2024 after 92% of its transaction volume went unmonitored

98.6% of Today's Reviews Find Nothing

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THE COST OF TODAY'S RULES-BASED REVIEWS



REVIEWS SENT BY THE RULES

11,750

235 days at 50 a day

ANALYST TIME SPENT

≈$290K

7,250 hours of review time

COST PER CASE FOUND

$1,778

The number the model has to beat

Jan–Aug 2023, 50 reviews a day · Assumes 37 minutes per review at $40/hour

MODEL OPERATIONS

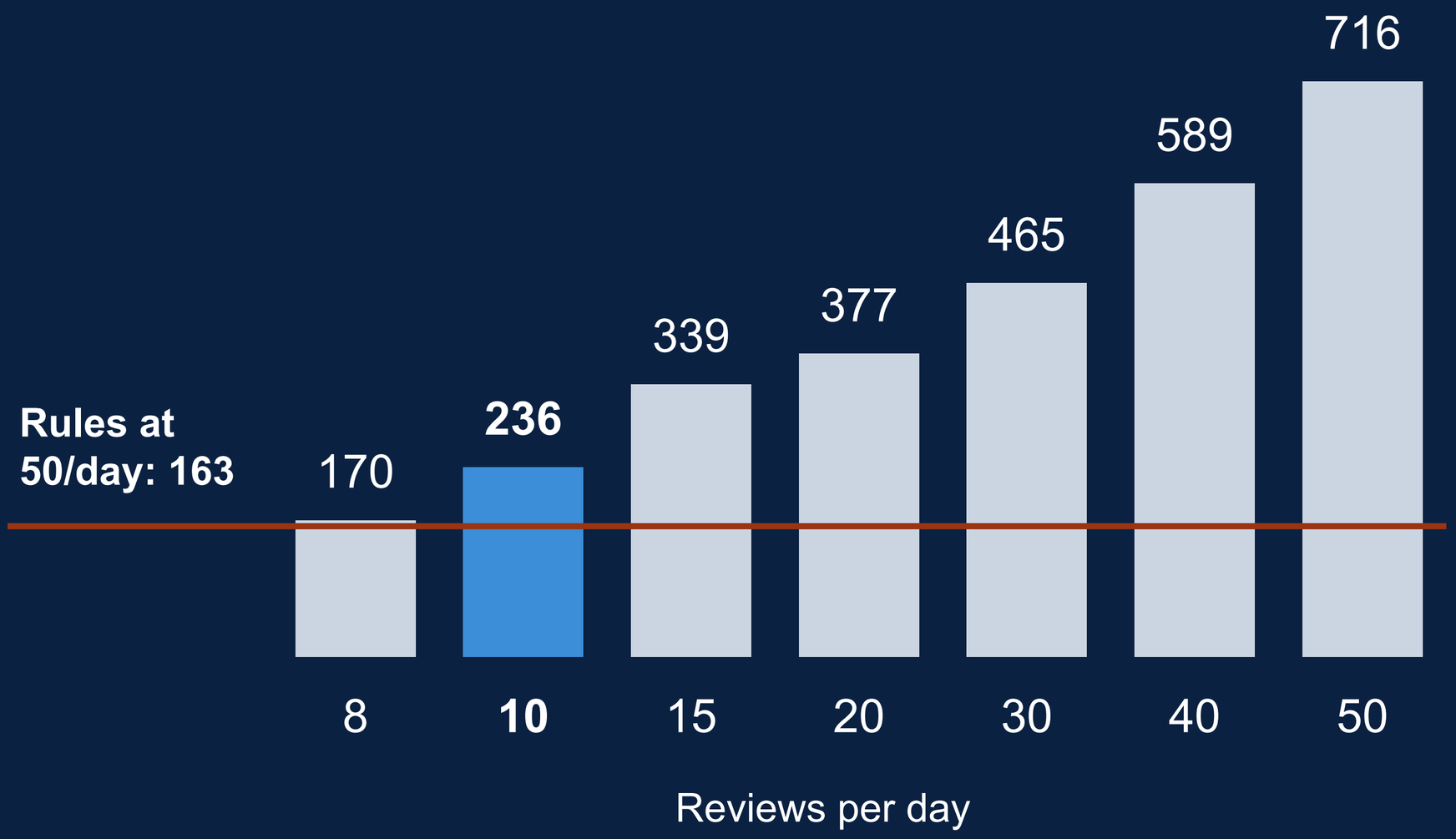
A Feedback Loop That Adapts Over Time

Incorporates investigator outcomes to retrain the model and maintain detection performance as behavior evolves.



Adaptive Alert Ranking Could Free Up to $360,333 Per Year

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REVIEWS AVOIDED

9,400

Compared with 50 reviews/day

ESTIMATED LABOR VALUE

$360,333

Potential capacity value

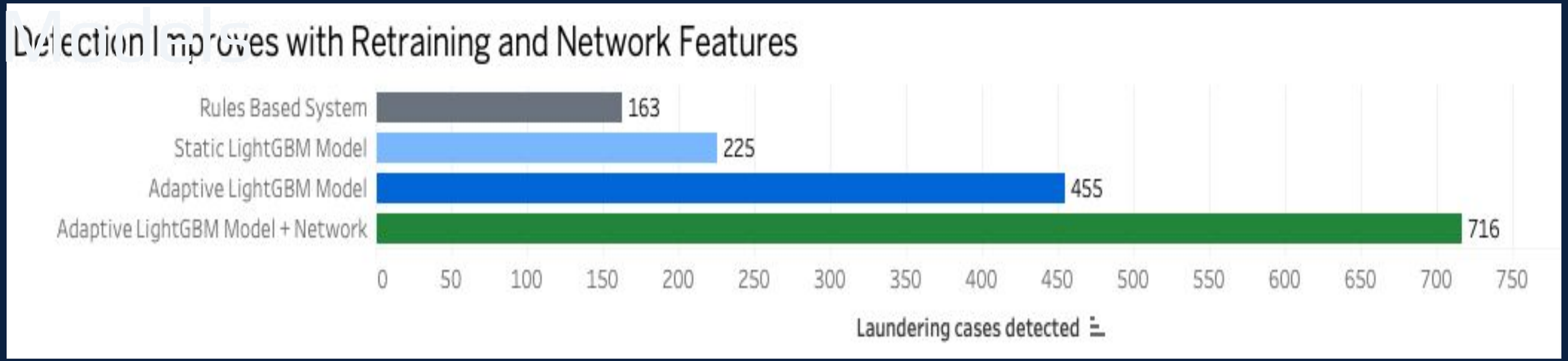
INVESTIGATOR HOURS FREED

5,797

At 37 minutes per review

Adaptive Alerts with Network Features Far Outperformed Other Models

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VS. TRADITIONAL RULES

4.4×

As many laundering cases detected

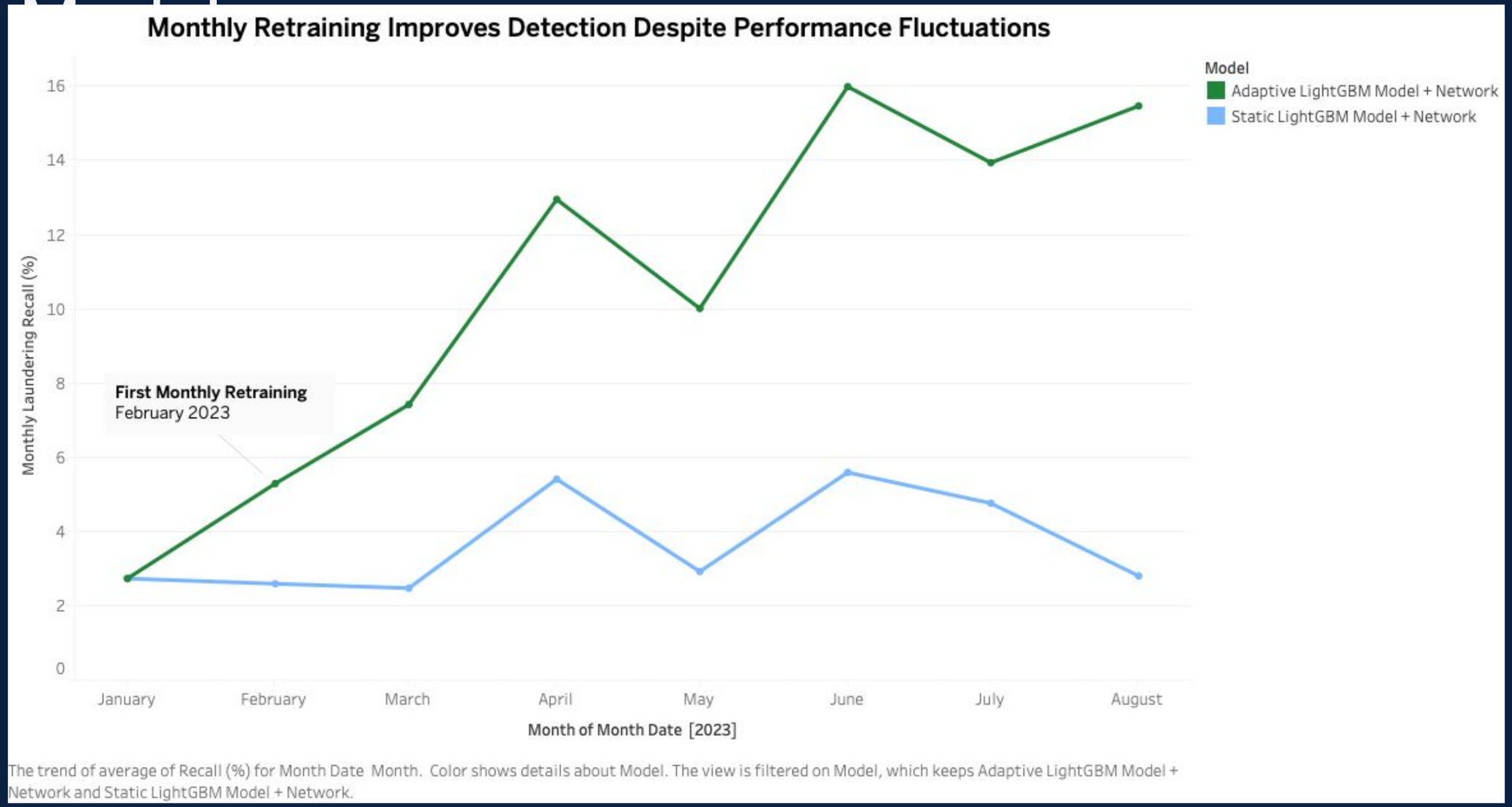
VS. ADAPTIVE MODEL

+57%

Additional cases detected with network features

Monthly Retraining Doubled Detection Without Increasing Reviews In The Same

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ADAPTIVE MODEL + NETWORK

716

CASES DETECTED

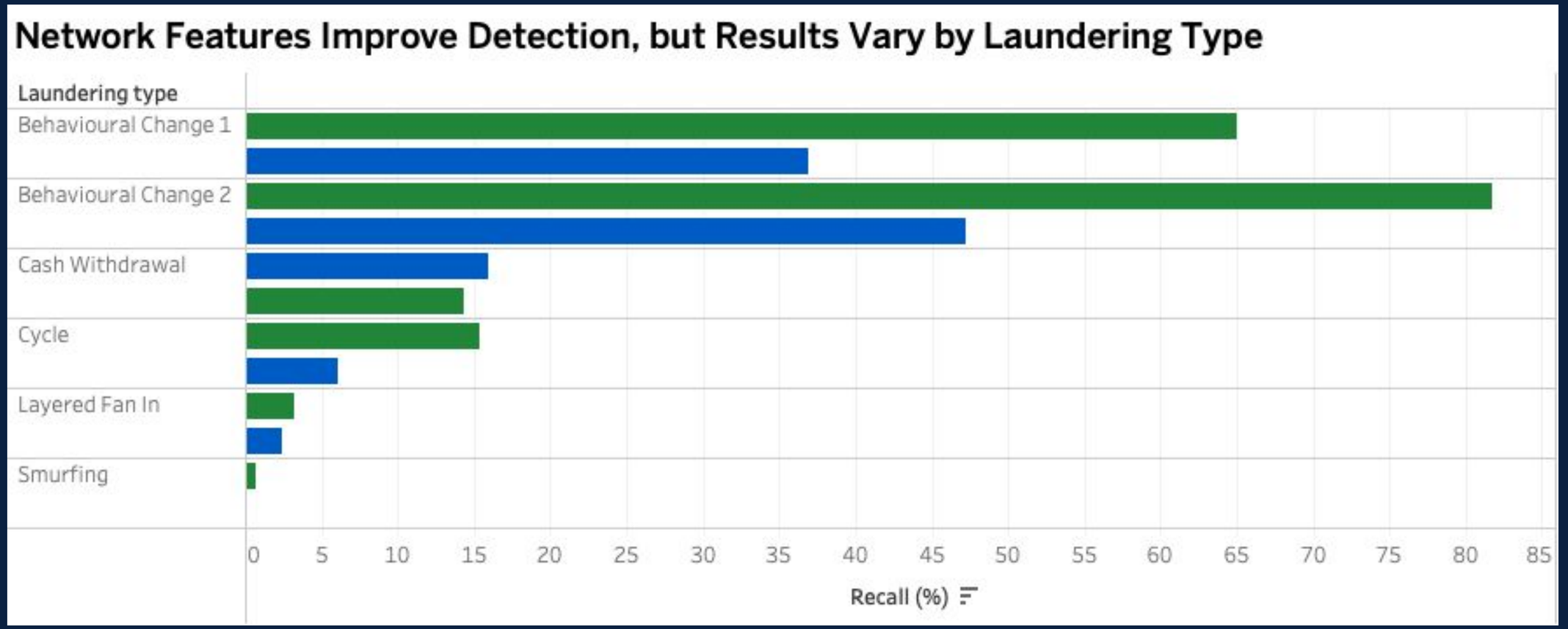
STATIC MODEL + NETWORK

254

CASES DETECTED

Network Context Improves Detection of Hidden Laundering Patterns

Rather than evaluating transactions in Isolation, the model examines how accounts interact, Identifying patterns of suspicious activity across connected accounts



How do we know these results are reliable?

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01

TIME BASED TESTING

Models were developed on earlier data and evaluated on later transactions to simulate forward looking performance

02

Equal Review Capacity

Every model was evaluated under the same review limit of 50 accounts per day over the 235 days.

03

Performance Across Time

Results were tracked monthly to evaluate whether detection improvements presisited as transaction patterns change

Thank You

QUESTIONS?

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Synthetic data; Kestrel Bank is fictional